AI101 Mini-Project · IIT Ropar

Laptop Price Prediction

Predicting a laptop's price from its specifications using regression

Team no. 85

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Repo link

github.com/2026aib1053-spark/laptop-price-predictor

Public repo: code, data, README, requirements, slides, video link

Problem and data

What and why

Predict a laptop's price (₹) from its specs. This is a regression problem.

Prices depend on many specs at once, so a model helps buyers judge if a price is fair and helps sellers price competitively.

Dataset

Source: [dataset name and link]

1,303 laptops × 11 columns, no missing values, no duplicates.

Target: **Price**, from ₹9,271 to ₹3,24,955 (right-skewed).

Columns

Company · TypeName · Inches · ScreenResolution · Cpu · Ram · Memory · Gpu · OpSys · Weight · **Price** (target)

Approach

1 Data
laptop_data.csv

2 Clean
RAM, weight, PPI, SSD/HDD, CPU family

3 Prepare
80/20 split, scale, one-hot encode

4 Model
Linear Regression (scikit-learn)

5 Output
Predicted price (₹) and metrics

Key steps

Predict the Price (₹) directly with Linear Regression.

Scale numbers (fit on train only) and one-hot encode categories.

Libraries used

pandas · numpy · scikit-learn · matplotlib · seaborn · Jupyter

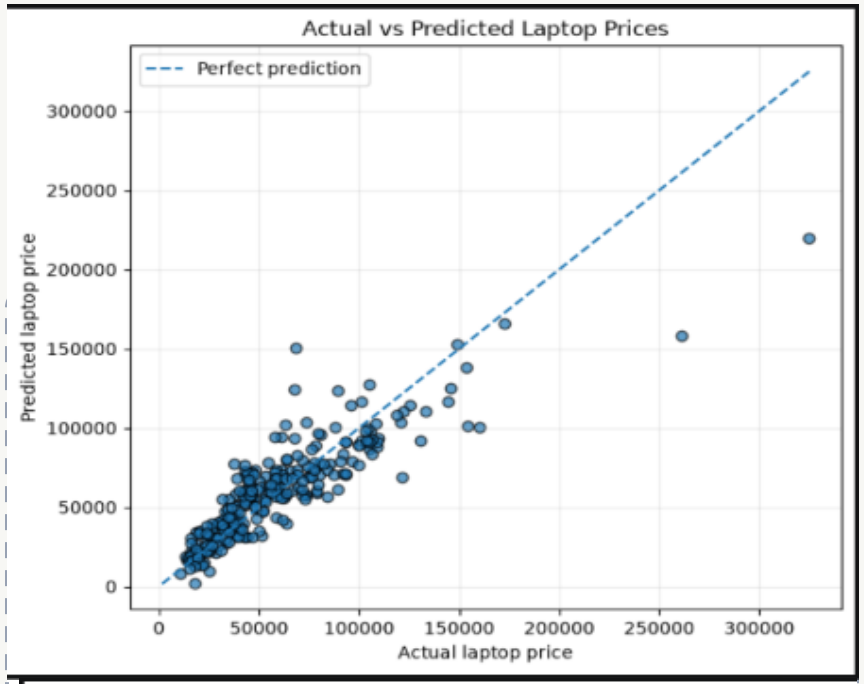
Results

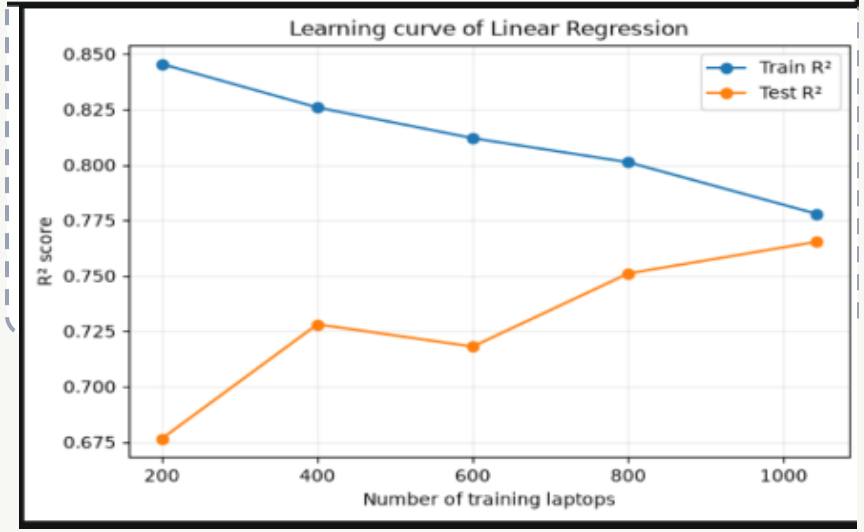
R^2 score (higher is better)

Train · 0.778

Test · 0.765

Test metric	Value
R^2	0.765
MAE	₹12,565
RMSE	₹18,397





Learnings and AI use

What we found

Linear Regression explains about 77% of price variation (R^2 0.77); train and test scores are close, so no overfitting.

RAM and SSD size have the strongest link with price (correlation heatmap).

Limits and next steps

Only 1,303 laptops from one time period; rare brands are predicted less reliably.

Next: newer data and other models (e.g. Random Forest).

AI-use disclosure

Claude: used for learning purpose and guidance in set up. It also help us in execution and understanding the format of Readme. Also used it making of ppt but mostly theory written by ppt lead head.